Groundstation capabilities
Concentric rings / Starlink

Listed antenna hardware / 3 November 2026 source

590 antenna records · 66 PoPs · 124 links

538 sized records · 52 unknown sizes · 91 incomplete inventories

Read the map

Triangle area scales with the largest complete listed antenna bank: max(N × π × D² / 4). Size uses antenna count and diameter; bands are never added together.

This is an aperture proxy for listed hardware, including planned and licensed installations. It is not throughput, antenna gain, installed capacity or a claim of current operation.

A hollow central square means size unknown. Small known triangles have a 0.04-inch source radius floor. Capability patterns are categorical: their rings, rays and petals are not antenna counts or RF beam diagrams.

Solid outline: at least one source operational flag is TRUE. Dashed: unconfirmed or planned. Dotted, muted outline: source says retired/withdrawn. Flags are not independently verified; raw status remains in the evidence receipt.

One point record per hex cell; triangles are centered inside the invisible cells. CDN grouping uses the same 0.0625-inch projected grid and hexoffsets routine. Cells fit the largest triangle in each group; a collision pass moves groups when needed. Tethers end at source positions.

PoPs remain small open circles. Fiber links retain supplied geometry, use the palette foreground at full opacity, and have no antenna symbol. Repeated coordinates are distinct source records, including upgrade proposals; no line endpoint is counted twice.

Sources / attribution

Unofficial Starlink Global Gateways & PoPs: user-supplied starlink-global-gateways-pops.20261103.kml and lossless-metadata GeoJSON conversion. The supplied file specifies neither author nor license. Antenna counts, bands and statuses come from its ExtendedData.

Diameter is interpreted in metres, consistent with FCC SpaceX 1.47 m and 1.85 m listings (DOC-374641A1; DOC-395259A1). This does not verify each site. Composite sizes and invalid fractional counts remain unresolved.

Starlink submission to Ofcom, 19 March 2025, pp. 4–5: existing V4 antennas support both Ka and E. This supports keeping band inventories separate; it does not validate the 2026 collection.

Natural Earth 1:10m land, public domain. Cartography and processing: Cartofreako. Atkinson Hyperlegible. Source URLs, raw fields, exceptions, formulas and hashes accompany the plates in capability-evidence.json and manifest.json.

Setouchi AuthaGraph: 43.5-inch image, +4.35-inch printed horizontal move, Antarctic terrestrial inset at lower left. Infrastructure keeps its geographic reference; display displacement is recorded separately.

Capability pattern / source record count

- Ka / 217
- Ka + E / 243
- E only / 9
- Q/V / 1
- Ka + E + Q/V + W / 66
- Community gateway / 21
- Tracking / telemetry / control / 4
- Ku / 1
- Unspecified / conflicting band / 28

Aperture proxy / square metres

13.6 · 50 · 100 · 200

Size samples print at the actual map scale.

- Source flag TRUE / 321
- Unconfirmed / planned / 250
- Retired / withdrawn / 19
- Unknown size / central square / 52
- PoP / fixed-size open circle / 66

493 projected source groups

Patterns enlarged above for comparison.

MSP speculation 2 / Ultramarine / porcelain

Randoma11y: ground #193fb8; marks/type/fiber links #f5eae7. sRGB contrast 7.28:1 (screen only). Fiber links use full opacity.
https://randoma11y.com/%23193fb8/%23f5eae7
Epson Stylus Pro 9900 / metallic glossy paper / cyan and light magenta reported low; grays reportedly print violet.
Intended RGB hues; final printed hue is unmeasured. No green correction, ink separation or metallic reflection simulation.